EXP. NO: 5

DATE: 13/2/2026

VERIFICATION OF NORTON'S THEOREM

AIM:

To verify norton's theorem practically and theoretically for the given DC circuit.

APPARATUS REQUIRED:

SL NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power supply (RPS)	(0-30V)	1
2.	Voltmeter	(0-30V) MC	1
3.	Ammeter	(0-10MA) (MC	1
4	Resistors	470 Ω 560 Ω 1K Ω	2 1 1
5.	Bread board	---	1
6.	multimeter	---	1

THEORY:

Norton's Theorem:

Norton's theorem states that, 'a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source, I_N ($= I_{SC}$) is parallel with a resistor R_N ($= R_{TH}$), where I_N ($= I_{SC}$) is the short-circuit current through the load terminal and R_N is the equivalent resistance at the load terminals when the independent source are turned off.

Norton's Current, I_N or I_{SC}:

It is the short-circuit current through the load terminals. i.e., $I_N = I_{SC}$

Norton's Resistance, R_N:

It is the look-back resistance across the load terminals when all the sources are replaced by their internal resistance. An ideal voltage source is replaced by short-circuiting as its internal resistance is zero. An ideal current source is replaced by open-circuiting as its internal resistance is infinity.

PROCEDURE :

1. Make the connections as per the circuit Diagram : 1

2. vary the RPS and set an input voltage of 10 V.

3. Note down the voltmeter reading (V_i) and ammeter reading (I_L) in Tabular column 1.

4. Switch off the supply and make connections for circuit Diagram 2.

5. Measure the Thevenin's resistance R_{TH} = Norton's resistance R_N.

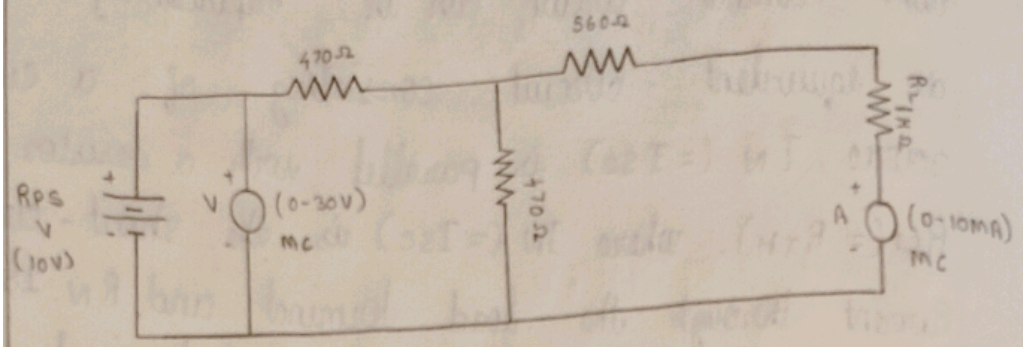
6. Switch off the supply and make connections for circuit Diagram : 3.

7. set an input voltage of 10 V in the RPS and note down the voltmeter readings V_i and V_{TH} ($= V_{oc}$) in Tabular column : 3

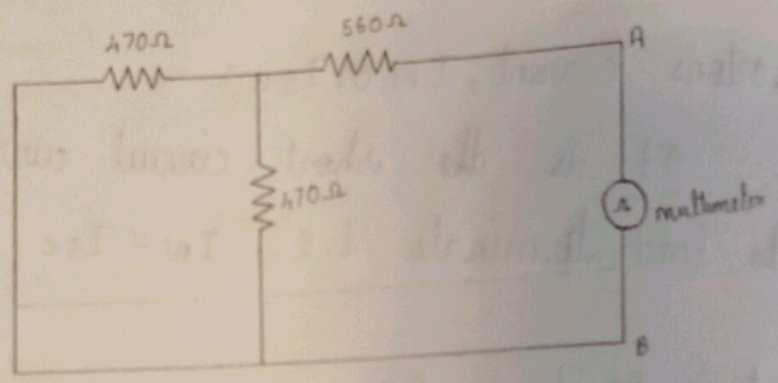
8. Switch off the supply and make connections for circuit Diagram 4.

CIRCUIT DIAGRAM: VERIFICATION OF NORTON'S THEOREM:

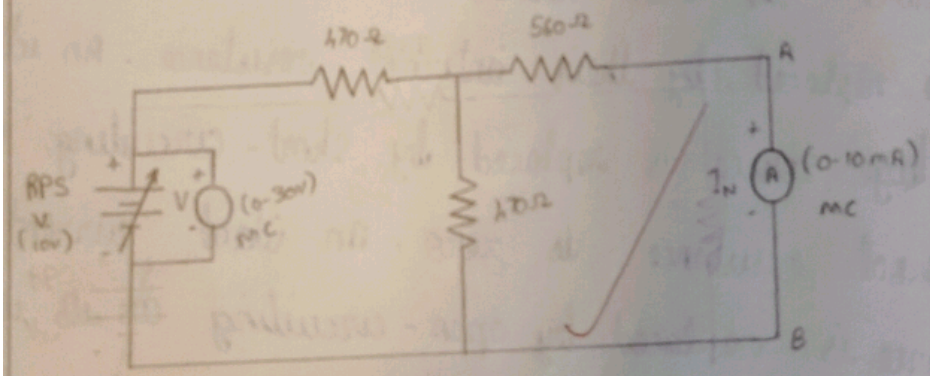
To measure I_L.



To measure R_{TH} or R_N



To measure I_N or I_{SC}



9. Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and Ammeter reading $I_N (= I_{SC})$ in tabular column 4.

10. Draw the Thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagram 5 & 6 respectively.

11. Calculate the I_L value using the formula

Thevenin's Theorem

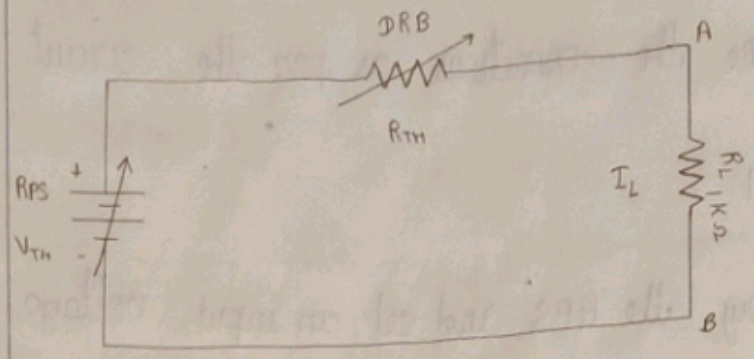
$$I_L = V_{TH}/(R_{TH} + R_L)$$

Norton's Theorem

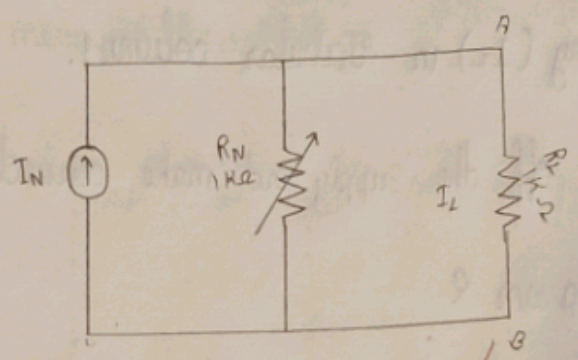
$$I_L = I_n * R_N / (R_N + R_L)$$

12. Theoretically verify the Norton's theorem.

Thevenin's equivalent circuit.



Norton's equivalent circuit:



TABULAR COLUMN: 1

To measure I_L

V_i (volts)	I_L (amps)
10V	2.75mA

TABULAR COLUMN: 2

To measure R_{TH} or R_N

V_i (volts)	R_{TH} (Ω)
10V	774

TABCLAR COLUMN: 3

To measure I_N or I_{sc}.

V_i (volts)	I_N (amps).
10 V	6.27mA

MARKS SPLIT-UP

SI.NO	Description	Marks Allotted	Marks obtained
1.	Experimental setup / connections	20 marks	20
2.	Observation / Readings	20 marks	20
3.	Calculations / Results	20 marks	18
4.	Viva	20 marks	18
5.	Record	20 marks	18
		Total.	94

RESULT:

Thus Thevenin's and Norton's theorem is verified practically and theoretically

Calculation

Theoretical calculation

short circuit the load resistance

measure In :

parallel

$R_{equ\,1} = 255.5\,\Omega$

$R_{eq\,2} = 235\,\Omega$

Total $= 255.5 + 470\,\Omega$

$= 725.5\,\Omega$

$$I_S = \frac{V}{R_I} = \frac{10}{25.5}$$

$= 0.013\,A$

$$I_N = 0.013 \times \frac{470}{470+560}$$

$I_N = 6.75\,mA$

To measure R_N :

$R_{equ} = 560 + 235$

$$R_N = R_2 + \frac{R_1 + R_3}{R_1 + R_3}$$

$= 560 + 235$

$= 795\,\Omega$

Load current :

$$I_L = I_N \cdot \frac{R_N}{R_N + R_L}$$

$$= 6.75 + \frac{795}{795+1000}$$

$= 2.768\,mA$

$V_L = 2.768\,V$